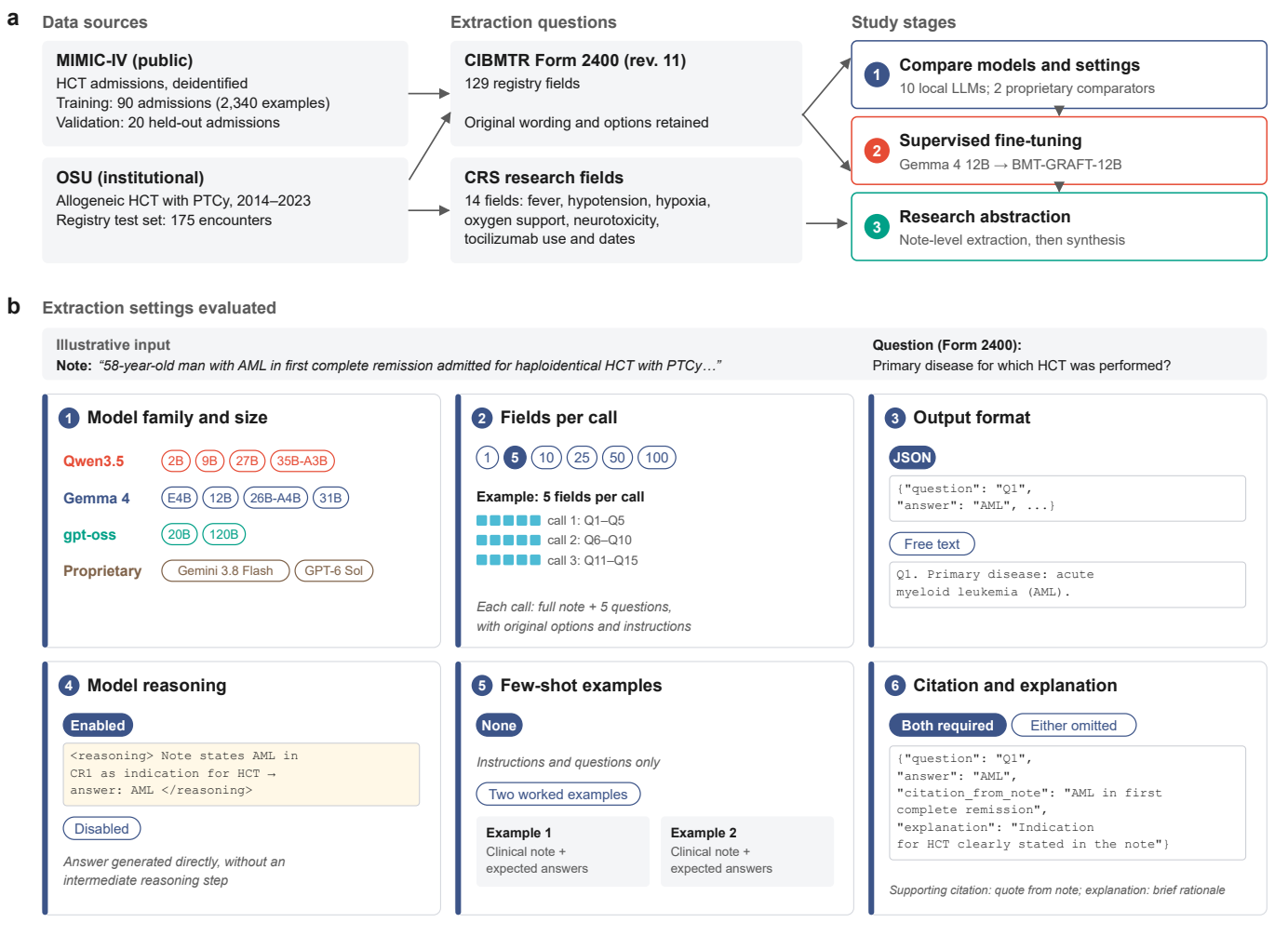

a
Data sources
MIMIC-IV (public)
HCT admissions, deidentified
Training: 90 admissions (2,340 examples)
Validation: 20 held-out admissions
OSU (institutional)
Allogeneic HCT with PTCy, 2014–2023
Registry test set: 175 encounters
Extraction questions
CIBMTR Form 2400 (rev. 11)
129 registry fields
Original wording and options retained
CRS research fields
14 fields: fever, hypotension, hypoxia, oxygen support, neurotoxicity, tocilizumab use and dates
Study stages
1 Compare models and settings
10 local LLMs; 2 proprietary comparators
2 Supervised fine-tuning
Gemma 4 12B → BMT-GRAFT-12B
3 Research abstraction
Note-level extraction, then synthesis
b
Extraction settings evaluated
Illustrative input
Note: "58-year-old man with AML in first complete remission admitted for haploidentical HCT with PTCy…"
Question (Form 2400):
Primary disease for which HCT was performed?
1 Model family and size
Qwen3.5
2B
9B
27B
35B-A3B
Gemma 4
E4B
12B
26B-A4B
31B
gpt-oss
20B
120B
Proprietary
Gemini 3.8 Flash
GPT-6 Sol
2 Fields per call
1
5
10
25
50
100
Example: 5 fields per call
call 1: Q1–Q5
call 2: Q6–Q10
call 3: Q11–Q15
Each call: full note + 5 questions, with original options and instructions
3 Output format
JSON
{"question": "Q1", "answer": "AML", ...}
Free text
Q1. Primary disease: acute myeloid leukemia (AML).
4 Model reasoning
Enabled
<reasoning> Note states AML in CR1 as indication for HCT → answer: AML </reasoning>
Disabled
Answer generated directly, without an intermediate reasoning step
5 Few-shot examples
None
Instructions and questions only
Two worked examples
Example 1
Clinical note + expected answers
Example 2
Clinical note + expected answers
6 Citation and explanation
Both required
Either omitted
{"question": "Q1", "answer": "AML", "citation_from_note": "AML in first complete remission", "explanation": "Indication for HCT clearly stated in the note"}
Supporting citation: quote from note; explanation: brief rationale